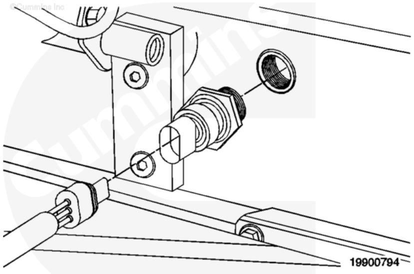
# Trainings

## Remove

Refer to Procedure 100-002 (Engine Diagrams) in Section E for sensor location information. (/qs3/pubsys2/xml/en/procedures/60/60-100-002.html)

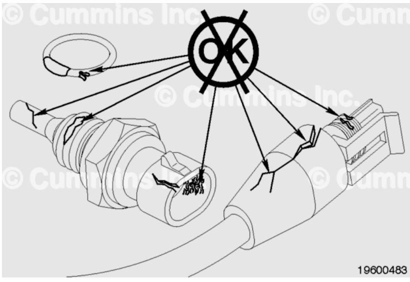



Disconnect the engine oil temperature sensor from the engine harness.

Remove the engine oil temperature sensor from the engine.

## Clean and Inspect for Reuse

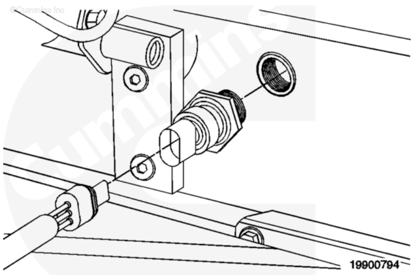
Inspect the engine harness connector and the engine oil temperature sensor for the following:




- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins
- Corroded, bent, pushed back, or expanded pins
- Damaged o-ring seal
- Thread damage.

## Install

Install a new o-ring onto the sensor, if necessary.



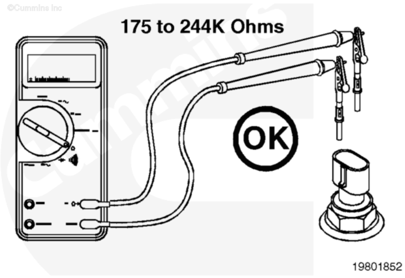

Install the engine oil temperature sensor.

**Torque Value:** 14 n•m [ 124 in-lb ]

Connect the sensor to the engine harness. A click will be heard when the connector locks in place.

## Resistance Check

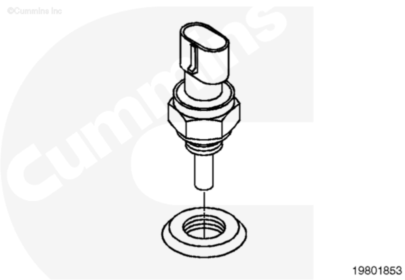
Disconnect the sensor connector. Select the resistance function on the multimeter. Attach the appropriate test leads to the multimeter probes. Touch the two test leads to the two terminals on the sensor. Measure the resistance. The multimeter **must** show between 115 and 244k ohms. The resistance value is temperature-dependent as shown in the table below.




| Temperature | Temperature | Range |
|---|---|---|
| (°C) | [°F] | (ohms) |
| 0 | 32 | 33k to 36k |
| 25 | 77 | 9k to 11k |
| 50 | 122 | 3k to 4k |
| 75 | 167 | 1350 to 1500 |
| 100 | 212 | 600 to 675 |

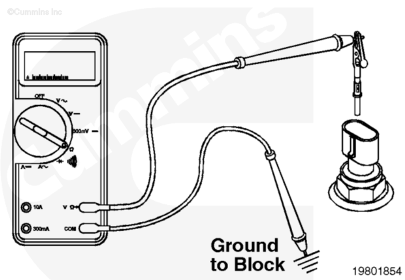
If the resistance is out of range, then the sensor has failed.

Replace the sensor.




## Check for Short Circuit to Ground

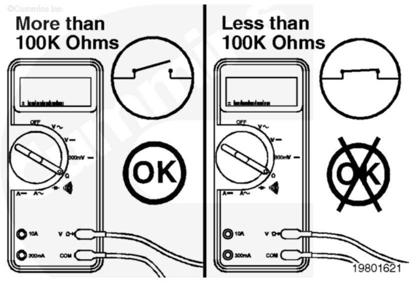
Touch the multimeter lead with the attached appropriate test lead to either terminal on the sensor. Touch the other multimeter lead to a good, clean surface on the engine block.




Measure the resistance.

The multimeter **must** show greater than 100k ohms, which is an open circuit. If the circuit is **not** closed, then there is a short within the sensor to chassis ground.

Replace the sensor.




**Last Modified: 14-Dec-2007**